```
#include <iostream>
using namespace std;

#define SIZE 5

class CircularQueue {
private:
   int queue[SIZE];
   int front, rear;

public:
   CircularQueue() {
      front = -1;
      rear = -1;
   }


   bool isFull() {
      return ((rear + 1) % SIZE == front);
   }

   bool isEmpty() {
      return (front == -1);
   }

   void enqueue(int value) {
      if (isFull()) {
         cout << "Queue is Full! Cannot insert " << value << endl;
         return;
      }

      if (isEmpty()) {
         front = rear = 0;
      }
      else {
         rear = (rear + 1) % SIZE;
      }

      queue[rear] = value;
      cout << value << " inserted into queue." << endl;
   }

   void dequeue() {
      if (isEmpty()) {
```

```
            cout << "Queue is Empty! Cannot delete." << endl;
            return;
        }

        cout << queue[front] << " deleted from queue." << endl;

        if (front == rear) {
            front = rear = -1;
        }
        else {
            front = (front + 1) % SIZE;
        }
    }

    void display() {
        if (isEmpty()) {
            cout << "Queue is Empty." << endl;
            return;
        }

        cout << "Queue Elements: ";

        int i = front;
        while (true) {
            cout << queue[i] << " ";

            if (i == rear)
                break;

            i = (i + 1) % SIZE;
        }

        cout << endl;
    }
};

int main() {
    CircularQueue cq;

    cq.enqueue(10);
    cq.enqueue(20);
    cq.enqueue(30);
    cq.enqueue(40);
    cq.enqueue(50);
```

```
    cq.display();

    cq.dequeue();
    cq.dequeue();

    cq.display();

    cq.enqueue(60);
    cq.enqueue(70);

    cq.display();

    cq.enqueue(80);

    return 0;
}
```